

Muhammad Waqar

+923086715563 | waqarsahi621@gmail.com | [linkedin.com/in/muhammad-waqar-1a594411a](https://www.linkedin.com/in/muhammad-waqar-1a594411a) | github.com/MuhammadWaqar621 | Islamabad, Pakistan

EDUCATION

University of Engineering and Technology (UET)

Bachelor of Science in Computer Engineering

Oct. 2015 – Aug. 2019

Taxila, Punjab

EXPERIENCE

Softoo

May 2025 – Present

AI Lead Engineer: Generative AI — RAG Systems — AI Agents

Islamabad, Punjab

- Leading AI agentic systems development and RAG (Retrieval-Augmented Generation) pipeline implementation for intelligent enterprise automation and decision-making platforms.
- Architecting multi-agent AI frameworks with autonomous reasoning capabilities, enabling complex task orchestration and self-improving system behaviors.
- Designing and deploying advanced RAG architectures with vector databases, semantic search, and context-aware knowledge retrieval for enterprise knowledge management systems.
- Specialized in local deployment of large language models including SQLCoder, Llama, and Code Llama for secure on-premise AI solutions, ensuring data privacy and reduced inference costs.
- Developed comprehensive AI-powered audit systems for automated compliance checking, risk assessment, and regulatory validation, streamlining audit processes through intelligent document analysis and anomaly detection.
- Mentoring junior engineers and establishing best practices for AI agentic development lifecycle and model governance in production environments.

Forbmax (pvt) Ltd

May 2023 – May 2025

AI Team Lead: Computer Vision — NLP — Data Science

Islamabad, Punjab

- Led a team of machine learning professionals, providing strategic guidance and technical mentorship to drive innovative solutions and collaborative efforts across multiple AI domains.
- Designed and developed advanced machine learning models for mission-critical applications, ensuring high performance and reliability with comprehensive testing and validation frameworks.
- Created and optimized end-to-end model pipelines, streamlining data preprocessing and model deployment processes, improving operational efficiency by 40% through automated workflows.
- Implemented custom training frameworks and fine-tuning strategies for transformer models, YOLO architectures, and computer vision models to achieve domain-specific performance improvements.
- Spearheaded fine-tuning of large language models and vision transformers using transfer learning techniques, achieving significant accuracy improvements on specialized datasets and business-specific use cases.
- Automated the annotation process through development of intelligent labeling systems and active learning frameworks, reducing manual annotation effort by 70% while maintaining data quality and consistency.
- Championed the design of robust APIs and microservices architecture, enhancing product accessibility and ensuring seamless integration with external systems and platforms for scalable AI solutions.
- Managed the successful deployment of machine learning solutions in production environments, optimizing real-time processing capabilities and reducing operational costs through efficient resource utilization.

Horizon Tech Services

Jun. 2020 – April 2023

Sr. AI Engineer

Islamabad, Punjab

- Served as a Senior AI Engineer, responsible for leading end-to-end machine learning projects while providing strategic guidance and mentorship to a dedicated team of professionals.
- Managed and supervised data creation, preprocessing, model training, and deployment for a variety of applications, ensuring data quality and model effectiveness.
- Oversaw extensive data handling, including data collection, cleaning, and annotation, to support the development of high-performing machine learning models.
- Designed and implemented robust data preprocessing pipelines, optimizing feature extraction and data augmentation techniques for enhanced model performance.
- Led the fine-tuning and optimization of models, with a particular focus on object detection, resulting in improved object recognition and tracking.

- Managed the deployment of machine learning solutions in production environments, collaborating cross-functionally to guarantee seamless integration and real-time processing.

Lampro Mellon IT Solutions Pvt. Ltd

Aug. 2019 – May 2020

Python Developer

Lahore, Punjab

- Worked as a Junior Python Developer, contributing to the development of software components with a focus on verification and high-level programming design.
- Collaborated with experienced team members to design and implement software modules, ensuring they meet quality and reliability standards.
- Assisted in the creation of automated verification scripts and tools to streamline the testing and validation processes.
- Participated in code reviews to learn and improve software quality and coding practices.
- Contributed to the design and architecture of software systems, applying theoretical knowledge gained during education.
- Worked on algorithm implementation and data processing, applying problem-solving skills to real-world challenges.

PROJECTS

Enterprise Compliance AI Agent Platform | *Agentic AI, RAG, LangChain, LangGraph*

Nov, 2025

- Designed and implemented an agentic AI platform that automates compliance checking across policies, contracts, logs, and internal documentation using multi-agent workflows.
- Built a production-grade RAG pipeline with vector stores and semantic search to ground LLM outputs in enterprise knowledge, improving factual accuracy and traceability of responses.
- Orchestrated tool-using AI agents with LangGraph and LangChain, enabling chained reasoning steps such as document retrieval, risk classification, redaction, and report generation.
- Deployed a hybrid LLM architecture combining self-hosted local models (e.g., Llama, DeepSeek) for sensitive data processing with Azure OpenAI/OpenAI APIs for high-quality generation where allowed.
- Implemented guardrails, audit logging, and policy-enforcement layers to ensure regulatory compliance, including prompt templates, response validation, and full decision traceability for auditors.

WellB - AI-Powered Hotel Management System | *RAG, Llama, OpenAI*

Feb. 2025

- Developed comprehensive AI chatbot “WellB” using RAG pipeline with local deployment of Llama and DeepSeek models, integrated with OpenAI GPT for enhanced conversational capabilities.
- Implemented intelligent hotel booking system with multi-model AI approach, enabling natural language booking management through locally deployed and cloud-based LLMs.
- Built emergency alert system with real-time notifications and intelligent priority classification using hybrid AI model architecture.
- Created AI-driven contact management with automated data extraction and contact addition using local Llama model deployment for data privacy.
- Developed booking edit functionality with conflict detection and automated rescheduling using DeepSeek model for complex reasoning tasks.
- Integrated multiple AI services creating cohesive WellB platform with local and cloud-based model orchestration for optimal performance and cost efficiency.

IntelliReport - AI-Powered Report Generation | *Python, NLP, Transformers, FastAPI*

Sep. 2024

- Developed comprehensive AI-powered report generation system with automated text correction, topic extraction, and intelligent summarization capabilities using transformer-based models and BERT architectures.
- Implemented advanced topic extraction engine utilizing BERT and custom NLP models to automatically identify and categorize key themes, subjects, and concepts from unstructured text data with hierarchical clustering.
- Built intelligent summary generation system using extractive and abstractive summarization techniques to create concise, coherent summaries while preserving critical information and maintaining contextual coherence.
- Created topic-wise summary functionality that segments documents by identified topics and generates targeted summaries for each category, enabling focused analysis and better content organization with semantic similarity scoring.
- Implemented smart text highlighting system that automatically identifies and emphasizes important sentences, key phrases, and critical data points using attention-based models and importance scoring algorithms with confidence thresholds.

- Deployed multi-format report generation with markdown and HTML rendering for direct frontend display and PDF download functionality for seamless user experience across different platforms and devices.
- Built scalable API architecture supporting batch processing with real-time progress tracking and efficient memory management for large document handling and concurrent user requests with load balancing capabilities.
- Integrated advanced analytics dashboard providing insights into document processing metrics, user engagement patterns, and system performance optimization recommendations for continuous improvement.

Smart Media Monitoring Application | *Python, PyTorch*

August 2023

- Led a comprehensive media monitoring project, involving the development and deployment of multiple AI models, including Speech to Text conversion, sentiment analysis, summary generation, topic modeling, facial recognition, and OCR.
- Worked with both Urdu and English datasets to ensure language versatility and provide valuable insights for the application's users.
- Developed a state-of-the-art Speech to Text model to transcribe spoken content into text, enhancing user accessibility and content indexing.
- Implemented sentiment analysis models to extract and categorize user sentiment from media content, enabling real-time monitoring of public opinion.
- Designed and optimized summary generation algorithms to distill lengthy media articles into concise, informative summaries, improving content consumption.
- Utilized topic modeling techniques to automatically categorize and organize media content, enhancing content discoverability and user experience.
- Pioneered facial recognition solutions to identify and track individuals in media content, providing valuable insights for security and analytics.
- Developed comprehensive text detection and OCR pipeline supporting multi-format document processing including PDFs, images, and tables with multi-language capabilities for Urdu, English, and Arabic text extraction.
- Implemented advanced OCR system with preprocessing techniques for noise reduction, skew correction, and layout analysis to handle complex document structures and varying image qualities.
- Integrated a YouTube Vlogger Scraper, automating data extraction for content analysis and audience insights.

Smart Surveillance System | *Python, PyTorch and TensorFlow*

December 2022

- Led the Smart Surveillance project focused on live camera streams and IP cameras, specializing in AI modules for comprehensive security monitoring.
- Live Object Detection: Implemented real-time object detection models for enhanced security and tracking with high accuracy and low latency performance.
- ANPR (Automatic Number Plate Recognition): Developed a highly accurate ANPR system for vehicle monitoring and identification with multi-language support.
- Geofencing: Created geofencing algorithms to define and enforce geographic boundaries within the camera streams for automated area monitoring.
- Video Summarization: Designed video summarization techniques to condense lengthy footage into concise and informative summaries for efficient review.

Interview Process Automation | *Python, FastAPI, PyTorch*

June 2022

- Led the Interview Process Automation project aimed at automating candidate selection by implementing various AI modules for comprehensive candidate assessment.
- Face Match: Developed a face recognition system to verify candidate identities with high accuracy and anti-spoofing capabilities.
- Face Consistency: Ensured facial consistency during different interview stages to prevent impersonation and maintain interview integrity.
- Voice Consistency: Implemented voice analysis to verify vocal consistency and prevent voice fraud using advanced audio processing techniques.
- Communication Skill Rating: Utilized AI to assess and rate candidate communication skills during interviews with natural language processing.
- Emotion Recognition: Implemented emotion recognition models to gauge candidate emotional responses during interviews for comprehensive behavioral analysis.

AAQAB Project | *Python, Yolo, PyTorch*

Feb. 2022

- Led the AAQAB Project, utilizing YOLO (You Only Look Once) and ResNet for object detection in large images with focus on satellite imagery analysis and geospatial intelligence applications.

- **Large Image Processing:** Worked on processing large images and splitting them based on latitude and longitude coordinates for efficient computational handling and distributed processing across multiple GPU clusters.
- **Object Detection with SAHI:** Implemented object detection algorithms with a focus on accurate object counting in high-resolution satellite imagery using sliding window approach for enhanced detection accuracy.
- **PRSS Image Pipeline:** Developed a comprehensive pipeline for PRSS (Pakistan Remote Sensing Satellite) image analysis with automated processing workflows and quality assurance mechanisms for consistent results.
- **XML to Text Conversion:** Automated the process of converting XML annotations to text format, facilitating data preprocessing for object detection models with validation and error handling capabilities.
- **YAML Generation for YOLO:** Automated the creation of YAML configuration files for YOLO model training, streamlining the training process with parameter optimization and hyperparameter tuning support.
- **Daily Report Generation:** Automated daily report generation to provide insights based on detected objects with statistical analysis, visualization dashboards, and trend monitoring for operational decision-making.
- **Performance Optimization:** Implemented GPU acceleration and memory optimization techniques to handle large-scale satellite imagery processing with reduced processing time and improved throughput efficiency.

Crowd Counting Automation | *Python, PyTorch*

June 2020

- Contributed to a project focused on crowd counting using computer vision techniques to address the challenge of accurately counting people in densely populated areas such as sports stadiums and large-scale festivals.
- **Crowd Counting Technique:** Collaborated in the development and implementation of crowd counting techniques to accurately determine the number of people in images using density estimation methods.
- **Dense Crowd Challenges:** Assisted in addressing the unique challenges of counting people in densely crowded scenarios, working towards innovative solutions for occlusion handling.
- **Efficiency and Speed:** Contributed to designing and optimizing the project for efficient crowd counting in large images with a focus on real-time processing capabilities.
- **Practical Implementation:** Played a role in translating research findings into practical solutions for automating crowd counting tasks, providing valuable insights in crowded environments.

Smart Meter Reader | *Python, FastAPI, PyTorch, Android*

August 2019

- Designed and developed the “Smart Meter Reader” as the final year project in computer engineering to automate the reading of utility meters using computer vision and deep learning techniques.
- **ROI Detection:** Implemented deep learning models for detecting Regions of Interest (ROI) within meter images, identifying two key areas - the meter digit area and the reference number.
- **Image Processing Pipeline:** Developed an image processing pipeline that cropped and isolated the selected ROIs for further analysis with noise reduction and enhancement techniques.
- **Reading Extraction:** Utilized deep learning models to extract readings from the selected ROI areas with high accuracy and error correction mechanisms.
- **Android Application Integration:** Integrated the project with an Android application that allowed users to upload meter images and receive the automated readings.
- **Cross-Network Access:** Ensured that the machine-learning model could be accessed from different networks, enabling remote usage and scalability.

TECHNICAL SKILLS

Programming Languages: Python

IDEs/Development Environments: VS Code, Jupyter Notebook, Google Colab, Anaconda, PyCharm

Libraries/Frameworks: TensorFlow, PyTorch, scikit-learn, Keras, OpenCV, Pandas, NumPy, FastAPI, Transformers

Databases: PostgreSQL, MySQL

Development Tools: Jira, GitHub, Git, Docker